

Objective-Driven Sequential Bayesian Experimental Design for Power-Grid Dynamic Learning and Control

[TODO: Authors and affiliations]

Abstract—This paper studies sequential Bayesian experimental design for uncertain power-grid dynamics. At each stage, a policy selects a probing bus and duration, observes a noisy frequency response, and updates uncertainty over machine-specific parameters. We distinguish parameter-oriented expected information gain (EIG) from Yoon’s mean objective cost of uncertainty (MOCU). We specialize the MOCU framework to a finite operational loss combining injection cost and unmet control requirement. Its Bayesian terminal decision is the posterior 95% quantile, and MOCU is expected regret relative to the model-specific optimum under that same loss. Random, fixed, myopic, deep adaptive design (DAD), and reinforcement-learning sequential Bayesian optimal experimental design (RL-sBOED), and Step-DAD form the comparison under common simulation banks and held-out protocols. A susceptible–infected–recovered (SIR) model provides an implementation benchmark; IEEE 9-bus, IEEE 14-bus, and IEEE 30-bus systems define the planned application sequence. IEEE30 dynamic implementation and the final multi-seed comparisons remain to be completed. [TODO: Add final quantitative findings.]

Index Terms—Bayesian experimental design, power-system dynamics, adaptive design, mean objective cost of uncertainty, information gain.

I. INTRODUCTION

UNCERTAINTY in power-system inertia and fast frequency response complicates frequency-security assessment and control selection. Controlled probing can reduce uncertainty, but an experiment must determine where to probe, how long to probe, and whether later probes should depend on earlier observations. [TODO: Add motivation and citations.]

EIG values parameter learning, but parameter information does not necessarily improve a final control decision. We therefore also study MOCU, which values an experiment by its reduction of expected decision penalty.

The study is organized around the following intended contributions:

- sequential bus–duration BOED for uncertain power-grid dynamics;
- a power-grid specialization of Yoon MOCU linking probing to the excess operational loss of a posterior-quantile control;
- a matched comparison of random, fixed, myopic, DAD, RL-sBOED, and Step-DAD; and
- SIR, IEEE9, IEEE14, and IEEE30 studies of objective choice, adaptation, horizon, and scalability.

[TODO: Funding and corresponding-author information.]

II. RELATED WORK

DAD amortizes sequential information-based design [1], RL-sBOED uses reinforcement learning [2], and Step-DAD adds history-specific online policy refinement [3]. MOCU instead values uncertainty through an operational decision and its loss [4], [5]. We specialize this decision framework to bus–duration probing, adapting the policy objectives accordingly. Controlled excitation motivates the frequency-response probes [6]. [TODO: Expand the comparison with related power-grid identification studies.]

III. PROBLEM FORMULATION

The formulation follows the physical and decision-making workflow: (i) define the grid dynamics and latent parameter space; (ii) define the reset probing experiment, design space, observation model, and posterior; (iii) define EIG for parameter learning; (iv) define the practical terminal control and safety standards; and (v) construct MOCU from the same control cost and safety quantities. Policy optimization is introduced only after these objects are fixed.

A. Power-Grid Dynamics and Latent Parameter Space

Let $\mathcal{N} = \{1, \dots, N\}$ denote all physical buses, and let $\mathcal{M} \subset \mathcal{N}$ denote the buses containing dynamic machines. For the reduced IEEE9 and IEEE14 models, the frequency response is modeled by

$$\mathbf{M}\dot{\boldsymbol{\omega}}(s) = -\Delta\mathbf{P}_e(s) - \left(\frac{\mathbf{K}}{2\pi} + \mathbf{D}\right)\boldsymbol{\omega}(s) + \mathbf{u}(s), \quad (1)$$

following the inertia–frequency-response model in [6]. The vectors $\boldsymbol{\omega}$ and $\Delta\mathbf{P}_e$ contain angular- frequency deviations and network electrical-power changes at the dynamic machines. The diagonal matrices $\mathbf{M}$, $\mathbf{K}$, and $\mathbf{D}$ contain inertia, fast frequency-response, and damping coefficients, respectively, and $\mathbf{u}$ is an external active-power injection.

In these reduced models, only the machine-specific coefficients are uncertain:

$$\boldsymbol{\theta} = ((M_i, K_i) : i \in \mathcal{M}) \in \Theta, \quad \boldsymbol{\theta} \sim p_0(\boldsymbol{\theta}). \quad (2)$$

Thus, every physical bus is an admissible probe location, whereas M_i and K_i are defined only for $i \in \mathcal{M}$. The IEEE9 and IEEE14 models have, respectively, $(N, |\mathcal{M}|) = (9, 3)$ and $(14, 5)$; their static network data follow the standard IEEE test cases distributed with MATPOWER [7].

Let $\mathbf{a}(b)$ map a scalar injection at physical bus $b \in \mathcal{N}$ to the dynamic-machine equations. At a machine bus, the injection enters that machine directly. At a non-machine bus, Kron reduction distributes the injection among the dynamic machines. A non-machine bus therefore remains a valid probe location without being assigned artificial dynamic states, M_i , or K_i .

B. Reset Probe Experiment and Design Space

At stage t , the design is a bus–duration pair

$$\xi_t = (b_t, \tau_t) \in \Xi = \mathcal{N} \times \mathcal{T}, \quad (3)$$

where $\mathcal{T}$ is the six-duration catalog and the amplitude A is fixed. The probing signal is the Hann pulse used for controlled frequency excitation in [6]:

$$u_p(s; \tau_t) = \begin{cases} \frac{A}{2} \left[1 - \cos\left(\frac{2\pi s}{\tau_t}\right) \right], & 0 \leq s \leq \tau_t, \\ 0, & s > \tau_t. \end{cases} \quad (4)$$

During a probe simulation,

$$\mathbf{u}(s) = \mathbf{a}(b_t)u_p(s; \tau_t), \quad u_{\text{ctrl}} = 0, \quad (5)$$

and the system is reset to its configured equilibrium before the next probe. IEEE9 and IEEE14 contain $9 \times 6 = 54$ and $14 \times 6 = 84$ actions, respectively. Exact repetition of a bus–duration pair is disallowed within a rollout.

After t probes, the realized history is

$$h_t = ((\xi_1, \mathbf{y}_1), \dots, (\xi_t, \mathbf{y}_t)). \quad (6)$$

An adaptive policy selects

$$\xi_t \sim \pi_t(\cdot \mid h_{t-1}), \quad t = 1, \dots, T, \quad (7)$$

subject to the no-repetition constraint. This is the standard adaptive formulation of sequential BOED [1]. The horizon T is the probe budget. A fixed design selects its entire sequence before any data are observed; an adaptive design may change later actions in response to the realized observations.

1) *Observation model and Bayesian update:* Let $o \in \mathcal{M}$ be the monitored PMU bus and $\Delta f_o(s) = \omega_o(s)/(2\pi)$ its frequency deviation. At prescribed sampling times $s_1, \dots, s_{N_{\text{obs}}}$, define the noise-free response

$$\mathbf{y}(\boldsymbol{\theta}, \xi_t) = [\Delta f_o(s_j; \boldsymbol{\theta}, \xi_t)]_{j=1}^{N_{\text{obs}}}. \quad (8)$$

The measured response is

$$\mathbf{y}_t = \mathbf{y}(\boldsymbol{\theta}^*, \xi_t) + \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \sigma_y^2 \mathbf{I}). \quad (9)$$

Here N_{obs} is a configurable observation dimension rather than a fixed part of the problem definition. Resetting and independent measurement noise give $p(\mathbf{y}_t \mid \boldsymbol{\theta}, h_{t-1}, \xi_t) = p(\mathbf{y}_t \mid \boldsymbol{\theta}, \xi_t)$. The adaptive selection probabilities depend on observed history, not additionally on the unknown parameter, so they cancel from the posterior:

$$p(\boldsymbol{\theta} \mid h_t) \propto p_0(\boldsymbol{\theta}) \prod_{k=1}^t p(\mathbf{y}_k \mid \boldsymbol{\theta}, \xi_k). \quad (10)$$

The implementation represents this posterior with N_p particles $\{\boldsymbol{\theta}^{(n)}, w_n^{(t)}\}_{n=1}^{N_p}$. Their weights are updated by

$$w_n^{(t)} = \frac{w_n^{(t-1)} p(\mathbf{y}_t \mid \boldsymbol{\theta}^{(n)}, \xi_t)}{\sum_{j=1}^{N_p} w_j^{(t-1)} p(\mathbf{y}_t \mid \boldsymbol{\theta}^{(j)}, \xi_t)}. \quad (11)$$

C. Expected Information Gain

Let H_T denote the random terminal history induced by policy π . For the parameter-learning experiments, the policy maximizes

$$\text{EIG}(\pi) = \mathbb{E}_{H_T \mid \pi} [D_{\text{KL}}(p(\boldsymbol{\theta} \mid H_T) \parallel p_0(\boldsymbol{\theta}))]. \quad (12)$$

This is the expected information gain used in sequential BOED [1], [2]. It rewards posterior learning about $\boldsymbol{\theta}$ but does not directly encode the quality of the downstream control decision.

D. Practical Control and Safety Conditions

Diagnostic probing and terminal control are evaluated in separate simulations. After the final probe, the operator selects a supplementary-injection magnitude $u \in \mathcal{U}$. Its location, activation time, and temporal profile are fixed. For model $\boldsymbol{\theta}$, contingency $c \in \mathcal{C}$, and magnitude u , the terminal simulator returns frequency nadir $f_{\text{nad}}(\boldsymbol{\theta}, c, u)$, maximum absolute ROCOF $r(\boldsymbol{\theta}, c, u)$. The intervention is safe for all credible contingencies when

$$\text{Safe}(\boldsymbol{\theta}, u) \iff \begin{cases} f_{\text{nad}}(\boldsymbol{\theta}, c, u) \geq f_{\text{lim}}, \\ r(\boldsymbol{\theta}, c, u) \leq r_{\text{lim}}, \end{cases} \quad \text{for every } c \in \mathcal{C}. \quad (13)$$

The next section expresses these requirements as a feasible control set and defines the minimum safe action. The retained parameter support must contain a safe candidate for every particle.

E. Finite-Loss Control-Oriented MOCU

1) *Relation to Yoon's MOCU:* For a specified operational loss $L(\boldsymbol{\theta}, u)$, the model-specific and intrinsically Bayesian robust (IBR) actions are, respectively,

$$u_{\boldsymbol{\theta}}^* \in \arg \min_{u \in \mathcal{U}} L(\boldsymbol{\theta}, u), \quad (14)$$

$$u_b^* \in \arg \min_{u \in \mathcal{U}} \mathbb{E}_b[L(\boldsymbol{\theta}, u)]. \quad (15)$$

Yoon et al.'s construction compares these actions using the same loss [4]; the generalized formulation also treats noisy experiments [5]:

$$\text{MOCU}(b) = \mathbb{E}_b[L(\boldsymbol{\theta}, u_b^*) - L(\boldsymbol{\theta}, u_{\boldsymbol{\theta}}^*)]. \quad (16)$$

Bayesian robustness means minimizing expected cost, not necessarily taking a worst-case maximum. The loss below is our power-grid specialization; neither its coefficient nor the 95% level is prescribed by Yoon et al.

2) *Physical control requirement and finite loss:* Let $\mathcal{U} = [0, u_{\max}]$ be the continuous terminal-control range. Define $U(\theta)$ as the minimum magnitude satisfying the prescribed nadir and ROCOF limits for every $c \in \mathcal{C}$. The scalar requirement model assumes that safety is monotone over this range and that a safe magnitude exists. These assumptions must be checked for each simulator and contingency set; otherwise a scalar threshold is insufficient and the full simulated action-dependent loss must be used. Equivalently, under these assumptions,

$$\mathcal{U}_{\text{safe}}(\theta) = [U(\theta), u_{\max}]. \quad (17)$$

The current IEEE9 scenario is the configured single contingency; a general set $\mathcal{C}$ does not establish N-1 security without evaluating that set.

For $\lambda > 1$ and $(z)_+ = \max(z, 0)$, choose

$$L(\theta, u) = u + \lambda[U(\theta) - u]_+, \quad \lambda = 20. \quad (18)$$

This finite loss penalizes supplementary injection and unmet required injection. It is a power-equivalent operational surrogate, not an empirically calibrated monetary damage model or a direct measure of violation severity. The coefficient is pre-specified to encode an asymmetric preference against under-control. A sensitivity study should vary this preference while changing the decision quantile consistently. No additional binary violation penalty or control margin is included in the primary formulation.

Since the loss decreases with slope $1 - \lambda$ below U and increases with slope 1 above U , its model-specific minimizer is

$$u_{\theta}^* = u_{\text{optimal}}(\theta) = U(\theta), \quad L(\theta, u_{\theta}^*) = U(\theta). \quad (19)$$

3) *Derivation of the 95% decision:* Write $F_b(v) = \Pr_b(U \leq v)$ and $\alpha = 1/\lambda = 0.05$. The left and right derivatives of posterior expected loss at u are $1 - \lambda + \lambda F_b(u^-)$ and $1 - \lambda + \lambda F_b(u)$, respectively. Convexity therefore gives the optimality condition

$$F_b(u^-) \leq 1 - \alpha \leq F_b(u). \quad (20)$$

Choosing the smallest minimizer yields

$$u_b^* = Q_{0.95}(U \mid b) = \inf\{v : F_b(v) \geq 0.95\}. \quad (21)$$

Thus the quantile is derived from the finite loss, rather than appended to an unrelated MOCU definition. Equivalently, realized regret is

$$\text{OCU}(\theta, b) = (u_b^* - U)_+ + 19(U - u_b^*)_+ \geq 0. \quad (22)$$

Over-control has unit marginal regret and under-control has marginal regret 19; the coefficient 20 in the total loss includes the avoided injection cost. The corresponding belief MOCU is

$$\text{MOCU}(b) = \mathbb{E}_b[u_b^* + 20(U - u_b^*)_+ - U]. \quad (23)$$

It vanishes when the required control is known exactly, even if other model parameters remain uncertain. Subtracting $\mathbb{E}_b[U]$ without including the shortfall term is not this finite-loss MOCU.

4) *Particle bank and implemented terminal action:* Let U_n denote the stored control requirement for particle $\theta^{(n)}$, with normalized posterior weights w_n . The empirical CDF and decision are

$$F_N(v) = \sum_n w_n \mathbf{1}\{U_n \leq v\}, \quad (24)$$

$$q_N = \inf\{v : F_N(v) \geq 0.95\}, \quad (25)$$

$$u_{\text{ctrl}}(b) = \text{snap}_{\mathcal{G}}^{\uparrow}(q_N). \quad (26)$$

Here $\mathcal{G}$ is the configured control grid. The aligned implementation requires every $U_n \in \mathcal{G}$ and zero margin, so the empirical quantile already lies on the grid and snapping does not change it. It is consequently an exact IBR action for this finite particle-bank loss. If requirements lie off-grid, rounding a continuous quantile upward is not generally the unconstrained grid Bayes optimum; one must minimize expected loss over the grid explicitly or retain continuous controls. The implementation checks this alignment before running the primary experiment.

The bank-level objective used for training and posterior evaluation is

$$\widehat{\text{MOCU}}_N(b) = \sum_n w_n \{u_{\text{ctrl}} + 20(U_n - u_{\text{ctrl}})_+ - U_n\}. \quad (27)$$

All methods use the same bank, loss, quantile, and control range. Probe and control bank rows must match in particle values and ordering, physics, scenario, units, and simulator provenance. A dense control-grid bank is a numerical approximation to the continuous physical requirements; its error must be distinguished from posterior uncertainty.

5) *Sequential design and held-out validation:* The ideal design objective and its implemented approximation are

$$\pi^* \in \arg \min_{\pi} \mathbb{E}_{H_T|\pi}[\text{MOCU}(b_T)], \quad (28)$$

$$\hat{\pi} \approx \arg \min_{\pi} \mathbb{E}_{H_T|\pi}[\widehat{\text{MOCU}}_N(b_T)]. \quad (29)$$

Under an exact posterior and the same generative model, the tower property equates expected posterior MOCU to expected realized regret. Also, $\mathbb{E}_{H_T}\mathbb{E}_{b_T}[U] = \mathbb{E}_{p_0}[U]$ is independent of the probe policy: minimizing expected MOCU is equivalent to minimizing expected total operational loss, not merely the control magnitude. Optimal expected MOCU cannot increase when additional observations may be ignored; this does not imply monotonic improvement for every realized history or trained policy.

For an independent test system θ^* , evaluate the applied action against a separately computed continuous physical oracle $U^*(\theta^*)$:

$$R^* = u_{\text{ctrl}} + 20[U^*(\theta^*) - u_{\text{ctrl}}]_+ - U^*(\theta^*). \quad (30)$$

Mean held-out R^* is the primary empirical endpoint; posterior bank MOCU is reported separately. Their difference can reflect finite support, control-grid approximation, and model or posterior miscalibration. Regret against a continuous oracle includes control discretization error and should not be described as purely uncertainty cost for a restricted grid action class. A strictly grid-relative MOCU instead subtracts $\min_{v \in \mathcal{G}} L(\theta, v)$ for each model.

TABLE I
METHOD CAPABILITIES; ADAPTIVE OR HORIZON-AWARE ARCHITECTURE
DOES NOT ESTABLISH A MEASURED PERFORMANCE ADVANTAGE.

Method	Adaptive	Non-myopic	Learned
Random	No	No	No
Fixed	No	Open-loop	No
Myopic	Yes	No	No
DAD	Yes	Yes	Yes
RL-sBOED	Yes	Yes	Yes
Step-DAD	Yes	Yes	Semi-amortized

Under the threshold model and exact posterior, the quantile ensures posterior coverage $\Pr_b(U \leq u_b^*) \geq 0.95$. It does not protect every posterior model and is not a guaranteed empirical safety rate. Report true-system nadir and ROCOF violations, safety counts and confidence intervals, control magnitude, and regret separately. Any 95% safety eligibility rule must be pre-specified; retain results for ineligible methods and distinguish a point estimate passing 95% from statistical evidence for that claim. The finite loss is scientifically a MOCU specialization, while its engineering adequacy depends on the stated loss preference, physical validation, and calibration.

IV. COMPARED DESIGN METHODS

Random ignores observations. Fixed selects an open-loop sequence. Myopic optimizes immediate expected utility after each observation. DAD learns an amortized history-conditioned policy offline. RL-sBOED learns a sequential policy with stepwise rewards. [TODO: Give reproducible implementation details.] For MOCU, the DAD-family loss is the finite-loss posterior objective, not the original information-gain objective. Discrete actions use score-function policy optimization. These are objective-based adaptations of the cited methods. Step-DAD [3] starts from an amortized DAD policy and performs limited history-specific online refinement. Report its offline DAD training cost, online refinement cost, and action-selection cost separately. [TODO: Describe the implementation and deviations from the original paper.]

V. EXPERIMENTAL SETUP

A. SIR EIG Benchmark

SIR verifies the EIG implementation on a simpler nonlinear model; it is not the application contribution. [TODO: State its parameters, actions, and noise.]

B. IEEE9 Experiments

IEEE9 has nine physical probe buses, three dynamic machines, and six uncertain M_i/K_i coordinates. Six durations give 54 actions. The main MOCU plan uses $N_{\text{obs}} = 5$, $\sigma = 0.005$, and $T \in \{3, 4, 5\}$. The current baseline catalog is $\{0.28, 0.32, 0.37, 0.40, 0.47, 0.62\}$ s; it is not asserted to be the final audited choice. Training seeds are 101, 202, and 303, crossed with evaluation seeds 1001–1005. A single training/evaluation pair is a pilot, not a complete publication cell. [TODO: Insert validated priors, final catalog, and completed-run provenance.]

C. IEEE14 MOCU Extension

IEEE14 has 14 physical probe buses, five dynamic machines, and ten uncertain M_i/K_i coordinates. Six durations give 84 actions. The MOCU workflow, methods, nominal noise, planned horizons, and seed protocol match IEEE9. Its priors, control scenario, thresholds, banks, and duration catalog require separate validation. [TODO: Insert the validated configuration and durations.]

D. IEEE30 MOCU Scalability Study

The planned IEEE30 extension uses the published dynamic reference of Demetriou et al. [8], not one dynamic machine per bus. The reference configuration has six machines: two governed generators and four synchronous condensers, with six added terminal buses (36 electrical buses in the modified network). The proposed uncertainty comprises six inertia constants and the two generator governor-droop parameters; the condensers are not assigned artificial governor coefficients. Prior ranges are study assumptions, separate from published nominal data. The full machine/exciter/governor backend is pending; this configuration cannot be treated as a completed reduced-swing experiment. [TODO: Validate initialization, dynamics, safety thresholds, probe mapping, master bank, and duration catalog before reporting IEEE30 results.]

E. Duration Selection and Evaluation

The IEEE9 master bank contains 281 durations from 0.20 to 3.00 s at 0.01 s spacing at each of nine buses; 2,529 candidate actions. A six-duration catalog includes all buses and therefore gives 54 actions. Full coverage of this dense grid is not exhaustive enumeration of the $\binom{281}{6}$ catalogs or a genuinely continuous simulator oracle. The current search uses global sampling, local refinement, and comparisons among calibrated Fixed, Myopic, and lookahead planners. It is independent of neural-policy training but is planner-based selection, so selection criteria and possible selection bias must be disclosed. Freeze finalists before fresh validation, account for multiple comparisons, and use a separate untouched test for final policy comparisons. Report the best catalog found, search coverage, and uncertainty; do not claim global optimality.

All methods share evaluation systems and noise streams. Final tables report one metric each, with methods as rows and T as columns. For trained methods, report mean $\pm$ sample standard deviation over the 3 training $\times$ 5 evaluation combinations, explicitly as descriptive crossed results rather than 15 independent training runs. Deduplicate non-training methods by evaluation seed (five results). Offline training/calibration time is summarized at run level, not repeated across evaluation seeds. Incomplete cells are em dashes; bold all tied best values in completed comparison columns. Paired uncertainty intervals resample independent test systems, keeping their noise replicas together; intervals conditional on a fixed bank do not quantify uncertainty from regenerating that bank. [TODO: Record bank sizes, split indices, audit budgets, oracle tolerances, run identifiers, and numbers of independent test systems.]

TABLE II
HELD-OUT REALIZED OPERATIONAL REGRET (MEAN $\pm$ SAMPLE STANDARD DEVIATION). TRAINED METHODS: 3 TRAINING $\times$ 5 EVALUATION SEEDS; NON-TRAINING METHODS: 5 UNIQUE EVALUATION SEEDS. EM DASHES DENOTE INCOMPLETE CELLS.

Method	$T = 3$	$T = 4$	$T = 5$
Random	—	—	—
Fixed	—	—	—
Myopic	—	—	—
DAD	—	—	—
RL-sBOED	—	—	—
Step-DAD	—	—	—

VI. RESULTS

A. SIR Implementation Validation

[TODO: Report EIG and runtime for every method.]

B. IEEE9 EIG Results

[TODO: Report mean $\pm$ standard deviation for $T \in \{3, 4, 5\}$.]

C. IEEE9 MOCU Results

[TODO: Report realized operational regret, posterior bank MOCU, safety, paired differences, and controls in separate tables.]

D. IEEE14 MOCU Results

[TODO: Report the same methods and metrics as IEEE9.]

E. IEEE30 MOCU Results

[TODO: Report the same methods and metrics as IEEE9 and IEEE14, emphasizing scalability in action count, latent dimension, training time, and online time.]

F. Computational Cost

[TODO: Separate bank generation, offline training, online selection, and Step-DAD history-specific refinement time.]

VII. DISCUSSION

Discuss when adaptation and non-myopic planning improve the downstream objective, why Fixed can be competitive, why EIG and MOCU can disagree, and how horizon and system size affect performance. Limitations include simulated Kron-reduced networks, prescribed priors, fixed probe amplitude, discrete actions, simplified terminal control, and three training seeds. Reduced-model K_i allocations and parameter priors are study assumptions. The paper-based IEEE30 model requires a different dynamic representation. The finite-loss coefficient needs sensitivity analysis and does not quantify actual outage damage; empirical calibration and physical safety must be validated separately. Adaptive branching and improvement over Myopic and Fixed are empirical hypotheses, not prerequisites imposed on the results.

VIII. CONCLUSION

This paper formulates power-grid probing as sequential BOED and connects learning to a finite-loss Bayesian terminal decision using Yoon’s MOCU construction. The 95% quantile follows from the chosen loss, while held-out safety and regret test its practical performance. [TODO: State only final conclusions supported by completed three-seed results.]

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[TODO: Add funding and computing resources.]

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